

Customer Shopping Behavior Analysis

Data Analytics Portfolio Project Report

Tools: Python | Pandas | PostgreSQL | SQL | Power BI

Dataset: 3,900 customer records, 18 attributes

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1. Introduction

1.1 Problem Statement

Retail businesses generate large amounts of customer transaction data, but raw data does not directly provide actionable insights. This project analyzes customer shopping behavior to understand purchasing patterns, customer spending, product performance, discount usage, subscription behavior, shipping preferences, and customer segments.

1.2 Objectives

- Clean and prepare the raw customer dataset for analysis using Python.
- Explore the data to understand its structure, distributions, and quality issues.
- Answer ten key business questions using SQL in PostgreSQL.
- Present the findings through an interactive Power BI dashboard.
- Translate the findings into practical business recommendations.

1.3 Tools and Technologies

Tool	Purpose
Python (Pandas)	Data loading, cleaning, feature engineering, and exploratory analysis
PostgreSQL / SQL	Business analysis queries, aggregations, and window functions
Power BI	Interactive dashboard and data visualization

2. Dataset Description

The dataset contains 3,900 customer records and 18 original attributes covering demographics, products, purchase amounts, reviews, discounts, subscriptions, shipping, previous purchases, payment methods, and purchase frequency. Each row represents one customer.

Column	Type	Description
Customer ID	int	Unique customer identifier (1 to 3,900)
Age	int	Customer age (18 to 70)
Gender	text	Male or Female
Item Purchased	text	Product bought (25 distinct items)
Category	text	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	int	Amount spent on the purchase
Location	text	Customer state
Size / Color / Season	text	Product size, color, and season
Review Rating	float	Customer rating of the product
Subscription Status	text	Whether the customer is subscribed

Column	Type	Description
Shipping Type	text	Six shipping options
Discount Applied / Promo Code Used	text	Yes or No
Previous Purchases	int	Count of earlier purchases (1 to 50)
Payment Method	text	Six payment methods
Frequency of Purchases	text	How often the customer buys

3. Exploratory Data Analysis and Cleaning (Python)

3.1 Dataset Structure

df.info() shows 3,900 rows and 18 columns: 4 integer columns, 1 float column, and 13 text columns. Every column is fully populated except Review Rating, which has 3,863 non-null values.

```
df.info()
```

```
PS C:\Users\ayush\OneDrive\Desktop\DataAnalytics> python app.py
PS C:\Users\ayush\OneDrive\Desktop\DataAnalytics> python app.py
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   object
3   Item Purchased        3900 non-null   object
4   Category              3900 non-null   object
5   Purchase Amount (USD) 3900 non-null   int64
6   Location              3900 non-null   object
7   Size                  3900 non-null   object
8   Color                 3900 non-null   object
9   Season                3900 non-null   object
10  Review Rating         3863 non-null   float64
11  Subscription Status    3900 non-null   object
12  Shipping Type         3900 non-null   object
13  Discount Applied      3900 non-null   object
14  Promo Code Used       3900 non-null   object
15  Previous Purchases    3900 non-null   int64
16  Payment Method        3900 non-null   object
17  Frequency of Purchases 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
```

Figure 1: Dataset structure from df.info()

3.2 Statistical Summary

The summary shows an average customer age of about 44 (range 18 to 70) and an average of about 25 previous purchases. Males make up 2,652 of 3,900 records, and 2,223 customers did not use a promo code. PayPal is the most common payment method (677 customers) and "Every 3 Months" the most common purchase frequency (584).

```
print(df.describe(include='all'))
```

	Customer ID	Age	Gender	Item Purchased	...	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	...	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	...	2	NaN	6	7
top	NaN	NaN	Male	Blouse	...	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	...	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	...	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	...	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	...	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	...	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	...	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	...	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	...	NaN	50.000000	NaN	NaN

Figure 2: Statistical summary of numerical and categorical columns

3.3 Missing Value Analysis

Only Review Rating contained missing values: 37 of 3,900 records (about 0.9%).

```
print(df.isnull().sum())
```

```
[11 rows x 18 columns]
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

Figure 3: Missing values before treatment (37 in Review Rating)

3.4 Missing Value Treatment

Missing ratings were filled with the median rating of the corresponding product category, which keeps the imputed values realistic for each type of product. A second null check confirmed that no missing values remained.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

```

dtype: int64
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD) 0
Location         0
Size            0
Color           0
Season           0
Review Rating    0
Subscription Status 0
Shipping Type    0
Discount Applied 0
Promo Code Used  0
Previous Purchases 0
Payment Method   0
Frequency of Purchases 0
dtype: int64

```

Figure 4: Missing values after treatment (all columns complete)

3.5 Feature Engineering and Transformation

Several changes were made to prepare the data for SQL and Power BI:

- Column standardization: column names were converted to lowercase snake_case.
- Age groups: customers were split into four equal-sized quartile groups (young adult, adult, middle age, senior) using `pd.qcut`.
- Purchase frequency: text frequencies were mapped to a numeric number of days (for example Weekly = 7, Fortnightly = 14, Annually = 365).
- Redundant column: `promo_code_used` was dropped because it was identical to `discount_applied` in every row (the equality check returned True).

```

df.columns = df.columns.str.lower().str.replace(' ', '_')
labels = ['young adult', 'adult', 'middle age', 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

print((df['discount_applied'] == df['promo_code_used']).all()) # True
df = df.drop(columns=['promo_code_used'])

```

```

Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount_usd', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
age    age_group
0    55    middle age
1    19  young adult
2    50    middle age
3    21  young adult
4    45    middle age
purchase_frequency_days  frequency_of_purchases
0                    14.0      Fortnightly
1                    14.0      Fortnightly
2                     7.0        Weekly
3                     7.0        Weekly
4                   365.0      Annually
True
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount_usd', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')

```

Figure 5: Standardized columns, age_group, purchase_frequency_days, and final column list

After cleaning, the dataset has 18 columns: 17 original columns (promo_code_used removed) plus age_group and purchase_frequency_days.

4. Business Analysis (SQL)

The cleaned dataset was loaded into PostgreSQL. The ten queries below answer key business questions. Result figures were verified by recomputing each query on the cleaned data.

4.1 Revenue by Gender

Compare total revenue generated by male and female customers.

```

SELECT gender, SUM(purchase_amount_usd) AS revenue
FROM customer_shopping_behavior
GROUP BY gender;

```

Result: Male customers generated \$157,890 and female customers \$75,191, so males contribute about 68% of revenue (and make up 68% of records).

4.2 Discount Users with Above-Average Spending

Identify customers who used a discount but still spent at least the overall average purchase amount.

```

SELECT customer_id, purchase_amount_usd
FROM customer_shopping_behavior
WHERE discount_applied = 'Yes'
AND purchase_amount_usd >= (
    SELECT AVG(purchase_amount_usd)

```

```
FROM customer_shopping_behavior);
```

Result: 839 customers used a discount and still spent at or above the average of \$59.76, so discounts are not only attracting low spenders.

4.3 Top 5 Products by Review Rating

Identify products with the highest average customer ratings.

```
SELECT item_purchased,  
       ROUND(AVG(review_rating)::numeric, 2) AS avg_rating  
FROM customer_shopping_behavior  
GROUP BY item_purchased  
ORDER BY AVG(review_rating) DESC  
LIMIT 5;
```

Result: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Handbag (3.78).

4.4 Standard vs Express Shipping

Compare average purchase amount between Standard and Express shipping customers.

```
SELECT shipping_type,  
       ROUND(AVG(purchase_amount_usd), 2) AS avg_purchase  
FROM customer_shopping_behavior  
WHERE shipping_type IN ('Express', 'Standard')  
GROUP BY shipping_type;
```

Result: Express customers spend slightly more on average (\$60.48) than Standard customers (\$58.46), a gap of about \$2.

4.5 Subscriber vs Non-Subscriber

Compare customer count, average spending, and total revenue by subscription status.

```
SELECT subscription_status,  
       COUNT(customer_id) AS total_customers,  
       AVG(purchase_amount_usd) AS average_spend,  
       SUM(purchase_amount_usd) AS total_revenue  
FROM customer_shopping_behavior  
GROUP BY subscription_status;
```

Result: See the table below. Subscribers spend about the same per purchase as non-subscribers, so subscription is not yet linked to higher spend.

Subscription	Customers	Avg. spend	Total revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

4.6 Products with Highest Discount Usage

Identify products where the highest share of purchases involved a discount.

```
SELECT item_purchased,  
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes'  
                             THEN 1 ELSE 0 END) / COUNT(*), 2) AS purchase_percentage  
FROM customer_shopping_behavior
```



```
GROUP BY item_purchased
ORDER BY purchase_percentage DESC
LIMIT 5;
```

Result: Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%). Overall, 1,677 of 3,900 purchases (43%) used a discount.

4.7 Customer Segmentation

Segment customers into New, Returning, and Loyal groups based on previous purchases.

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer_shopping_behavior)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type GROUP BY customer_segment;
```

Result: Loyal: 3,116 (80%), Returning: 701 (18%), New: 83 (2%). The customer base is dominated by repeat buyers.

4.8 Top 3 Products in Each Category

Identify the three most purchased products within every category using a window function.

```
WITH item_counts AS (
  SELECT category, item_purchased,
    COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category
      ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer_shopping_behavior
  GROUP BY category, item_purchased)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3;
```

Result: See the table below. Ties are broken arbitrarily by ROW_NUMBER().

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a (only 2 items)

4.9 Repeat Buyers and Subscription

Analyze subscription status among customers with more than five previous purchases.

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer_shopping_behavior
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Result: Of 3,476 repeat buyers, only 958 (28%) are subscribers and 2,518 (72%) are not. Many loyal customers have not subscribed.

4.10 Revenue by Age Group

Determine which age group contributes the most revenue.

```
SELECT age_group, SUM(purchase_amount_usd) AS total_revenue
FROM customer_shopping_behavior
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Result: See the table below. Revenue is spread fairly evenly across the four groups.

Age group	Age range	Customers	Total revenue
Young adult	18 to 31	1,028	\$62,143
Middle age	45 to 57	986	\$59,197
Adult	32 to 44	942	\$55,978
Senior	58 to 70	944	\$55,763

Age ranges are approximate quartile boundaries (quartile cut points at ages 31, 44, and 57).

5. Power BI Dashboard

The cleaned data was visualized in an interactive Power BI dashboard titled "Customer Shopping Behavior Analysis".



Figure 6: Customer Shopping Behavior Analysis dashboard (Power BI)

5.1 Dashboard Components

Component	What it shows
KPI cards	3.9K customers, \$59.76 average purchase amount, 3.75 average review rating
% of Subscribers	Donut chart: 27% subscribed, 73% not subscribed
Revenue by Category	Clothing leads, followed by Accessories, Footwear, and Outerwear
Sales by Category	Order counts follow the same ranking as revenue
Revenue by Age Group	Horizontal bar chart of revenue per age group
Sales by Age Group	Horizontal bar chart of order counts per age group
Slicers	Subscription Status, Gender, Category, and Shipping Type

5.2 Key Dashboard Insights

- Clothing is the top category, with \$104,264 in revenue from 1,737 purchases. Accessories is second at \$74,200.
- Only 27% of customers are subscribers, which leaves a large group to target.
- Purchase counts are close across age groups (942 to 1,028), with young adults the highest.

Note on the age-group revenue chart: the bars in the current dashboard show values of roughly 80K to 90K per group, which would total far more than the dataset revenue of \$233,081. The SQL result in section 4.10 (about \$56K to \$62K per group) is the figure consistent with the data. It is worth rechecking the measure and sort order used in that visual before publishing the dashboard.

6. Key Findings

- The dataset is clean after treatment: only 37 ratings (0.9%) were missing and these were imputed by category median.
- Revenue totals \$233,081 across 3,900 customers, with an average purchase of \$59.76 and average rating of 3.75.
- Males account for about 68% of revenue (\$157,890).
- Clothing and Accessories together generate about 76% of revenue.
- 43% of purchases used a discount, and several products (Hat, Sneakers, Coat) have close to 50% discount usage.
- Subscribers spend about the same as non-subscribers (\$59.49 vs \$59.87).
- 80% of customers are Loyal (more than 10 previous purchases), but only 28% of repeat buyers are subscribed.

7. Business Recommendations

- **Increase subscription adoption:** Only 27% of customers subscribe, even among repeat buyers. Offer targeted benefits such as free shipping or early access to convert loyal non-subscribers.
- **Make subscription more valuable:** Subscribers currently spend no more than others. Add perks that encourage larger baskets.

- **Target discounts instead of applying them broadly:** 839 customers spent above average even with a discount, and 43% of purchases are discounted. Use personalized offers for high-value customers.
- **Monitor discount dependency:** Hats, Sneakers, and Coats are discounted in about half of purchases. Review whether these products need discounts to sell.
- **Focus on high-revenue categories:** Prioritize inventory and marketing for Clothing and Accessories, which generate most revenue.
- **Build loyalty programs:** Loyal and Returning customers make up 98% of the base. Reward them to protect repeat revenue.
- **Grow the female customer segment:** Female customers generate less than half the revenue of males. Test targeted campaigns and product ranges.
- **Use segments for marketing:** Combine age group, gender, and customer segment to design targeted campaigns.

8. Conclusion

This project covered the full analytics workflow: cleaning and preparing the data in Python, answering business questions in SQL, and communicating results in a Power BI dashboard. The analysis shows a loyal, repeat-buying customer base concentrated in Clothing and Accessories, heavy use of discounts, and low subscription uptake. The recommendations focus on converting loyal customers to subscribers, using discounts more selectively, and focusing on the highest-revenue categories.

Future Scope

- Add customer lifetime value and churn analysis.
- Analyze seasonality using the season and purchase frequency fields.
- Compare performance by location (state).
- Build a predictive model for subscription or high-spend customers.